

COMPUTER ORGANIZATION PRACTICE SHEET

NIMCET TARGET PAPER • COMPUTER BASICS

ResourceOS

1. How many $32K \times 1$ RAM chips are needed to construct a memory system with a total capacity of 256 Kbytes?

- (A) 16
- (B) 32
- (C) 64
- (D) 128

2. If a processor clock is rated at 2500 MHz, what is the clock period of this processor?

- (A) 2.50×10^{-10} sec
- (B) 4.00×10^{-10} sec
- (C) 1.00×10^{-10} sec
- (D) 5.00×10^{-10} sec

3. A system has a main memory access time of 100 ns and a cache access time of 10 ns. If the hit ratio is 0.90, what is the effective memory access time?

- (A) 19 ns
- (B) 20 ns
- (C) 18.5 ns
- (D) 28 ns

4. In byte-addressable memory, the memory address range from 0000H to 3FFFH represents a total memory size of:

- (A) 4 KB
- (B) 8 KB
- (C) 16 KB
- (D) 32 KB

5. How many 2-to-4 line decoders with an enable input are required to construct a 4-to-16 line decoder?

- (A) 4
- (B) 5
- (C) 6
- (D) 7

6. A memory unit has a 16-bit address bus and 16-bit data bus. If $1K \times 8$ RAM chips are used to construct this memory, how many chips are required?

- (A) 64
- (B) 128
- (C) 256
- (D) 512

7. What is the theoretical maximum speedup ratio of an N-stage instruction pipeline executing a large number of tasks ($M \gg N$) over a non-pipelined processor?

- (A) $N - 1$
- (B) N
- (C) $N / (N + 1)$
- (D) N^2

8. Which unit mapping correctly identifies the standard transfer units for CPU, Cache, and Main Memory respectively?

- (A) CPU → Block, Cache → Word, Memory → Page
- (B) CPU → Word, Cache → Block, Memory → Page
- (C) CPU → Page, Cache → Word, Memory → Block
- (D) CPU → Word, Cache → Page, Memory → Block

9. If a computer instruction format specifies 16 distinct opcodes and 32 general-purpose registers, what is the minimum length of the control field needed to specify 2 source registers and 1 opcode?

- (A) 12 bits
- (B) 14 bits
- (C) 16 bits
- (D) 18 bits

10. A 4-stage pipeline processes 100 instructions. If each stage takes 2 ns clock cycle time, what is the total time taken to complete all 100 instructions?

- (A) 200 ns
- (B) 206 ns
- (C) 800 ns
- (D) 208 ns

11. Consider the following statements regarding the CPU Fetch-Decode-Execute cycle:

I. The Program Counter is incremented after an instruction is fetched.

II. The ALU decodes the control signals to determine the operation type.

Which of the above statement(s) is/are correct?

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

12. What is the start address and end address range for a 2 KB memory module if the starting address assigned is 2000H?

- (A) 2000H to 27FFH
- (B) 2000H to 2FFFH
- (C) 2000H to 37FFH
- (D) 2000H to 23FFH

13. In a system with a 32-bit address space and 4 KB page size, the page table contains entries for mapping. How many bits are allocated for the Page Number?

- (A) 10 bits
- (B) 12 bits
- (C) 20 bits
- (D) 22 bits

14. In a microprogrammed control unit, if the control memory has 512 words of 32 bits each, how many bits are required for the Control Memory Address Register (CMAR)?

- (A) 8 bits
- (B) 9 bits
- (C) 10 bits
- (D) 32 bits

15. In an instruction format with zero addresses (0-address instruction machine), operands are implicitly accessed from:

- (A) Accumulator
- (B) General Purpose Registers
- (C) Top of Stack
- (D) Main Memory Cache

16. A computer uses a 20-bit address bus to access memory. If the memory is word-addressable and each word is 32 bits (4 bytes), what is the total memory capacity in bytes?

- (A) 1 MB
- (B) 2 MB
- (C) 4 MB
- (D) 8 MB

17. Which statement accurately compares hardwired control units and microprogrammed control units?

- (A) Hardwired is slower but more flexible than Microprogrammed
- (B) Hardwired is faster and less flexible than Microprogrammed
- (C) Microprogrammed is faster and easier to modify than Hardwired
- (D) Microprogrammed uses combinational circuits while Hardwired uses ROM

18. Which CPU register stores the status flags such as Zero (Z), Carry (C), Sign (S), and Overflow (V)?

- (A) Program Counter
- (B) Memory Buffer Register
- (C) Program Status Word / Flag Register
- (D) Instruction Register

19. A memory chip is specified as 2048×8 . What is the number of data output lines and address input lines required for this chip?

- (A) 8 data, 11 address (B) 11 data, 8 address
(C) 8 data, 10 address (D) 16 data, 11 address

20. Which data transfer scheme uses handshaking signals (e.g., STROBE or ACK) to synchronize data transfer between asynchronous devices?

- (A) Synchronous DMA (B) Asynchronous Data Transfer
(C) Programmed Memory Bus (D) Serial Interrupt Cycle

21. In a DMA transfer, when the DMA controller takes control of the system bus from the CPU for one memory cycle to transfer a single word, it is called:

- (A) Burst Mode (B) Cycle Stealing
(C) Interrupt Request Mode (D) Polling Mode

22. A 16-bit register contains the binary value 1001 1100 0101 0000. What is its representation in Hexadecimal?

- (A) 9C50H (B) 9D50H
(C) 8C50H (D) 9C40H

23. How many memory references are required to fetch and execute a 1-address indirect addressing mode instruction?

- (A) 1 (B) 2
(C) 3 (D) 0

24. Which memory organization type allows simultaneous read/write operations by maintaining separate read and write data buses?

- (A) Single-port RAM (B) Dual-port RAM
(C) Serial EEPROM (D) Dynamic RAM

25. In a computer system, the register that keeps track of the location of the next microinstruction in control memory is:

- (A) Program Counter (PC) (B) Control Address Register (CAR)
(C) Instruction Register (IR) (D) Memory Address Register (MAR)

ANSWER KEY

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(C)	6	(B)	11	(A)	16	(C)	21	(B)
2	(B)	7	(B)	12	(A)	17	(B)	22	(A)
3	(A)	8	(B)	13	(C)	18	(C)	23	(C)
4	(C)	9	(B)	14	(B)	19	(A)	24	(B)
5	(B)	10	(B)	15	(C)	20	(B)	25	(B)